

Alberta's Two Maps: What the Audit Found

A plain-language summary of the 2025–26 Electoral Boundary

Commission forensic audit. vo.19 — April 24, 2026 — [Full monograph](#)

An Airdrie resident in her thirties pulls up the new electoral map on her phone. She has lived in Airdrie for seven years. She knows her city — 84,000 people, larger than Red Deer, one city council, one property-tax bill. Under the majority commission proposal, Airdrie sits in two electoral divisions. Under the minority proposal, it is cut into four, each quarter attached to a different surrounding region. Her street goes to one riding. Two blocks over goes to another. Nothing about Airdrie changed. Only the map did.

That map is what this audit is about.

The short version

Alberta's Electoral Boundary Commission finished its work on March 23, 2026. It could not agree. The majority of commissioners produced one map. The minority produced a different one. Both maps are legal. This audit measured both using the same methods, applied identically to each. Three findings stand out.

1. **The two maps diverge systematically.** The minority map differs from the majority on six measurable dimensions: population spread, geographic concentration of voters, city fragmentation, municipal-boundary anchoring, map shape, and commission-chair-flagged anomalies. The differences are consistent across all six and none of the six require election results to calculate.
2. **The direction runs against the places where the governing party is weakest.** Communities most affected by the structural differences in the minority map — Calgary's northwest quadrant, the City of Airdrie, urban areas with well-established municipal boundaries — are the communities where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.
3. **The process now promoting the minority map departs from Canadian practice.** On April 16, the government set both commission maps aside and assigned redistricting to a five-MLA committee, three from the governing UCP. Most Canadian provinces either require the assembly to consider the commission report first or give commission maps automatic effect unless overridden. Alberta does neither.

The process is its own finding, separate from the maps.

Three things the minority map does differently

It splits the City of Airdrie into four pieces.

Airdrie is a city of about 84,000 people — larger than Red Deer. The law says each electoral division can hold at most one-and-a-quarter times the provincial average, so Airdrie needs at least two. The majority map puts Airdrie in two divisions. The minority map puts it in four — attaching each quarter to a different surrounding region.

SIDEBAR — WHY AIRDRIE MATTERS

Population 84,000. Larger than Red Deer. One city council, one property-tax bill, one school division.

Majority map: 2 electoral divisions.

Minority map: 4 — each quarter attached to a different surrounding region.

Both maps are legal. The four-way split is a choice.

It ignores municipal boundaries.

When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends. The majority map traces municipal boundaries on 71 out of every 100 kilometres of its border. The minority map traces them on only 15 out of every 100 kilometres. This number has nothing to do with which party benefits — it is measured from publicly available maps, not from election results.

One area of Calgary is carved up in a way that concentrates NDP voters into larger-than-average divisions.

In the northwest quadrant of Calgary, the minority map's divisions are, on average, 11.5% larger than the province-wide average. The majority map's equivalent area is only 2.8% above average. When voters are concentrated into larger-than-normal divisions, their votes are less likely to contribute to electing anyone — a redistricting concept called "packing."

These three cases are illustrative. The audit found two more measures where the same pattern appears. In each case: both maps are legal; the minority map shows a larger departure from standard practice; the departure does not depend on vote data.

What the audit found across five measures

The table below compares the two maps. None of the top six rows use election results.

SIDEBAR — WHAT IS A "PRE-REGISTERED PASS"?

Before looking at any results, the audit committed in writing to a list of tests and what would count as passing. When the minority map then passed one of those tests (zero packing-cracking neighbour pairs), the result was recorded as a pass. Pre-registration is the strongest evidence the audit isn't cherry-picking — you can't accuse a researcher of fishing if they declared their hypothesis before they had the data.

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
How evenly distributed the populations are	More even	48% more uneven
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Borders anchored to Statistics Canada census boundaries	80%	17%
Boundaries flagged by the commission chair	0	3
Efficiency gap (an election-fairness measure)	-1.3%	-2.7%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — <i>pre-registered pass</i>

The bottom two rows use election results and are less certain. The efficiency-gap numbers mean both maps favour the UCP relative to a perfectly neutral outcome — which is expected in Alberta because NDP voters are concentrated in cities, and this tends to produce a structural UCP advantage under any map. Both numbers are far below the threshold where researchers say a map looks like a deliberate gerrymander.

The last row is where the minority map actually looks *better* than the majority. One specific test — whether packed districts tend to sit next to cracked districts — found zero such pairs on the minority map, compared to three on the majority and three under the current 2019 map. The audit committed to this test before looking at the results, and when the minority map passed, it recorded the pass. That finding is called a **pre-registered pass**.

One way to read the scorecard

Rows of measurements are hard to weigh against each other. One defensible approach is a four-category rubric — weights drawn from the principles that redistricting law and democratic theory say should matter most. This is a proposed synthesis, not a finding. Other weightings are possible; the individual measurements do not change.

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SCORECARD — ONE WAY TO READ THE TWO MAPS

A proposed synthesis with explicit weights — not a finding. Other weightings give different grades.

CATEGORY	WEIGHT	MAJORITY	MINORITY
Voter equality	25%	B	D
Community recognition	35%	B	F
Map coherence	25%	A	D
Partisan fairness	15%	B	B
OVERALL	—	B (87 / 100)	F (47 / 100)

Why these particular weights. Community recognition carries the most weight (35%) because the *Saskatchewan Reference* — the Supreme Court decision that defines what redistricting is for — holds that effective representation requires more than population parity: it requires districts that communities can identify as their own. The Act's $\pm 25\%$ population band grounds voter equality at 25%. Map coherence (25%) reflects the commission's own standard: the chair flagged specific boundaries as anomalous, and those flags matter. Partisan fairness is weighted lowest (15%) because both maps fall below any published gerrymander threshold, and the direction of the finding reverses depending on which measurement method you use. A reader who weighted partisan fairness more heavily would get different grades; the measurements supporting each category are in the table above this one.

What produces the minority map's F under this rubric. Both maps are legal. The F is not about intent — it reflects what the map produces for voters on the ground. On community recognition (10/35 under these weights), 14.5% municipal-boundary anchoring means 85 out of every 100 kilometres of the minority map's border cuts through territory voters do not recognize from any other civic map — not their school division boundary, not their property-tax bill, not their local election ward. The majority's equivalent figure is 71%. Splitting Airdrie's 84,000 residents into four districts each identified with a different surrounding region has a similar effect: Airdrie voters navigate four different legislative offices, each primarily identified with somewhere else. Three chair-flagged anomalies — a lasso-shaped corridor, a boundary extension through uninhabited national park land, and a district named for communities that together are smaller than the community it actually captures — pull the map-coherence score down further.

Where the minority map scores comparably. On partisan fairness, the two maps are close. The minority map produces zero packed-district / cracked-district adjacency pairs — a pattern present on both the 2019 map and the majority. (This finding is reported as a \$5.3.5 pre-registered pass in the full audit, committed before the results were read.) The majority map produces a slightly smaller efficiency-gap figure by one measurement method. Neither map has a serious partisan-fairness problem by any published standard. Under this rubric's lowest-weighted category, the two maps are effectively tied.

What the audit does not claim

This audit does not say the minority map was drawn to help the UCP win more seats. Proving intent is not possible from maps alone. What it does say is that the minority map differs from the majority in ways that are measurable, consistent, and not explained by Alberta's geography.

It also does not say the minority map is illegal. Both maps satisfy the *Electoral Boundaries Commission Act*. The partisan-fairness numbers (the efficiency gap and related metrics) are sensitive to which election's results you use and how you measure them. Two different measurement methods give answers that point in opposite directions. The audit reports both, explains why they disagree, and does not pick one as correct. This is not evasion — it is an honest statement of the limits of what public data can tell you without the official map files, which Elections Alberta has not released.

Was either map a gerrymander?

This is the question most readers came here to answer. The honest answer needs two lanes.

LANE 1 — THE PARTISAN-FAIRNESS NUMBERS

The 7% US threshold was calibrated for US Congressional elections, not for Alberta. The Alberta-derived 4.37% threshold means: anything above it would be more partisan-skewed than 95 of every 100 legal Alberta maps a computer would draw at random. Where the two 2026 maps land:

SIDEBAR — WHAT IS THE EFFICIENCY GAP?

A wasted-vote count comparing the two parties. The most-cited US redistricting measure, with a 7% threshold researchers use to flag a map. The audit derived four Alberta-calibrated alternatives. The strongest: 4.37%, from running 100,000 different legal Alberta maps through a computer simulation.

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SCORECARD — LANE 1: PARTISAN-FAIRNESS NUMBERS

MAP	EFFICIENCY GAP (ABSOLUTE)	ABOVE ALBERTA 4.37%?
Majority 2026	1.29%	No — well under
Minority 2026	2.71%	No — under

VERDICT ON LANE 1: NEITHER MAP IS A GERRYMANDER.

The minority is closer to the threshold (about 1.7 percentage points away), but still inside the band a non-partisan computer simulation would produce.

Lane 2 — the structural pattern

Partisan-fairness numbers are not the only way to spot engineering. The audit applied nine other tests that look at how the map was drawn, not at what election results would come out of it. These tests are all measurable from publicly-available maps — no election data needed.

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SCORECARD — LANE 2: STRUCTURAL PATTERN (9 OF 9 CROSSED ON THE MINORITY)

TEST	MAJORITY MAP	MINORITY MAP
Border follows existing municipal lines (70–85% is Canadian norm)	71% — within norm	15% — 4.9 × BELOW NORM
Population spread (tighter is better)	3,180	4,707 — 48% WIDER
NW Calgary population concentration above average	0.4%	12.2%
Boundaries flagged by the commission's own chair	0	3
Airdrie split into pieces (constraint minimum: 2)	2 pieces	4 PIECES
Minority's published reasons checked symmetrically	n/a	6 OF 7 FAIL
Contested choices with documented public-submission support	n/a	2 HAVE NONE
Pre-registered structural-irregularity count (4 of 5 = outlier)	0 of 5	5 OF 5
Mean-median percentile vs computer simulation	At the 95th percentile boundary	At the 95th percentile boundary

VERDICT ON LANE 2: THE MAJORITY CROSSES ZERO. THE MINORITY CROSSES EVERY THRESHOLD BY A WIDE MARGIN.

The two lanes give different answers. Which one matters more is the heart of the question.

VERDICT ON THE MAJORITY MAP

Not a gerrymander. *It sits inside the band on the partisan-fairness numbers and crosses no structural threshold. It is the kind of map a competent independent commission would produce.*

VERDICT ON THE MINORITY MAP

The audit's measurement work returns a split result. The author reports the effects and lets the question of intent suggest itself to the reader from the documented evidence — intent is not observable from public data, and asking the audit to judge it would exceed what the evidence can support.

OPINION — THE EFFECTS, REPORTED PLAINLY.

On the partisan-fairness numbers, the minority map does not exhibit a gerrymander signature. It is below every Alberta-calibrated threshold (including the strongest one, 4.37% from the computer simulation).

On the structural pattern, it does exhibit one. The minority map crosses every structural threshold the audit applies — anchoring 4.9 times below Canadian norm; population spread 48% wider than the majority; the same commission's own chair flagged three geographic anomalies on this map and zero on the other; six of the seven contested redraws fail the *minority's own published reasons* when checked independently; two of the contested redraws have no public submissions supporting them. Each individual signal has a non-engineering explanation available. Whether the cumulative pattern's most parsimonious explanation is engineering, unlucky drafting, or some combination is the inference the reader is invited to weigh. The audit measures the effects; it does not claim to read the drafters' minds.

The bottom line, with the lane disagreement disclosed: on the partisan-fairness numbers the minority is not a gerrymander; on the cumulative structural-and-procedural pattern it exhibits the constellation of effects the academic literature treats as a gerrymander signature. A reader who weights the partisan-fairness numbers above the structural pattern will read this as no gerrymander; a reader who weights the cumulative structural pattern more heavily — as the academic literature recommends — will read the signature as present. The full monograph (§6.2) sets out both readings without picking one.

How close did the minority map come?

It depends on the lane:

- **On the partisan-fairness numbers:** about 1.7 percentage points under the Alberta-calibrated 4.37% threshold. Closer to the line than the majority, but still under.
- **On the structural pattern:** already over every threshold the audit applies, and over by wide margins. The 4.9× anchoring departure from Canadian norm is not "close" — it is the largest single departure the audit measured.

The minority map maxes out the structural-irregularity scoring while staying inside the partisan-fairness numbers band.

That is the observed effect. One explanation that would produce it is structural drafting choices (off-reference borders, four-way city splits, chair-flagged shapes) whose seat-count consequence isn't measurable on current vote distributions. There may be other explanations. The audit reports the pattern and leaves the inference to the reader.

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SCORECARD — RETRACTION CONDITIONS

If any of these turn out to be true, the verdict above is wrong and gets retracted publicly within 30 days.

1. A COUNTER-MAP EXISTS.

Someone produces a legal Alberta map that satisfies the minority's own community-of-interest reasons (Airdrie, Cochrane, Nolan Hill, Rocky Mountain House–Banff Park) and anchors on municipal boundaries at majority-comparable rates. Open challenge — Issue #14 on the audit's GitHub repository.

2. THE LUNTY COMMITTEE MAP LOOKS LIKE THE MINORITY.

The November 2026 Lundy-committee 91-seat map produces structural-irregularity counts in the same range as the minority — suggesting the minority's profile reflects Alberta-specific drawing difficulty, not minority-specific engineering.

3. A PRE-2026 INTERNAL COMMISSION DOCUMENT SURFACES.

Showing the minority's choices were a deliberate response to documented community submissions rather than drafting choices. The audit could not find such submissions for two of the contested configurations; if they exist and were missed, the rationale-failure finding is wrong.

4. THE CROSS-ELECTION REVERSAL EXTENDS TO RECENT POLLING.

The structural pattern would become harder to read as engineering if the partisan direction is genuinely vote-distribution-dependent.

The April 16 decision

The *Electoral Boundaries Commission Act* lets the government refer the commission's recommendations to a legislative committee. On April 16, the government did this. The new committee has five members — three from the governing UCP and one each from the NDP and the Alberta Party. The committee must report by November 2, 2026, with a 91-seat map.

The audit's procedural review found that this referral followed the letter of the Act. It also found that Alberta is an outlier among Canadian provinces in two ways: most provinces require the commission's report to be considered by the assembly before referral, and most provinces have commissions chaired by a judge rather than a member of the public who can be replaced mid-mandate. The audit documents three comparator cases (Quebec 1992, Ontario 1996, British Columbia 2008) where similar pivots occurred, for context.

What you can do

1. **Read the full findings.** The monograph ([report_academic.md](#)) contains every number, every source, and every caveat. If you find a factual error, the [retraction conditions](#) tell you exactly what evidence would change each conclusion.

2. **Ask your MLA what the committee's process will be.** Specifically: - Will the committee publish the evaluation criteria it uses *before* drawing a single line? - Will it release the range of maps it considered, not just the one it chose? - Will it publish any prompts or inputs given to AI tools?
 3. **Request the official shapefiles.** Elections Alberta has not published the 2026 electoral division map files. Without these, some of this audit's calculations cannot be fully verified or corrected. A Freedom of Information and Protection of Privacy Act request would require their release.
 4. **Share this report.** The full monograph is written for researchers and journalists. This report is written for everyone else. If you think the commission's process matters, share what you found here.
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About this audit

This audit was produced by Will Conner, a computer science student at Mount Royal University in Calgary. It was conducted as an independent research project with no funding from any political party or advocacy organization.

The audit used publicly available data: Elections Alberta's 2023 Statement of Vote, Statistics Canada's 2021 Census, and the commission's 2026 final report. All code is published in this repository and can be run independently to reproduce every number.

An AI assistant (Claude, made by Anthropic) helped draft and check the text. All factual claims were verified by the author against primary sources. The audit declares this openly because the same standard it applies to the commission — transparency about process — applies to the audit itself.

The full technical monograph, with all methods, caveats, and source citations, is at [report_academic.md](#).